

1. Given the equation $x^n + y^n = z^n$ for x, y, z and n positive integers.

(a) (5 points) For what values of n is the statement in the previous question true?

(b) (2 1/2 points) For $n = 2$ there's a theorem with a special name. What's that name?

(c) (2 1/2 points) What famous mathematician had an elegant proof for this theorem but there was not enough space in the margin to write it down?

Total for Question 1: 10

2. (20 points) Compute

$$\int_0^{\infty} \frac{\sin(x)}{x}$$